PIMPRI CHINCHWAD EDUCATION TRUST's
PIMPRI CHINCHWAD COLLEGE OF ENGINEERING
(An Autonomous Institute)

S.Y. B. TECH **Year:** 2025 – 26

Semester: IV

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Department: Information Technology **Division:** A

Course: Database Management System Laboratory **Course Code:** BIT24PC04

Assignment No.2

Problem Statement: SQL Queries- Use Create, Select, Update, delete with operators, functions, and set operator:
a. Design and Develop SQL DDL statements which demonstrate the use of SQL objects such as Table, Index and different constraints etc.
b. Write at least 10 SQL queries on the suitable database application using SQL DML Statements.

Enter password: ******
Welcome to the MySQL monitor. Commands end with ; or \g. Your
MySQL connection id is 23
Server version: 8.0.45 MySQL Community Server - GPL Copyright (c)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

#Command to display the list of all databases present in the MYSQL server.

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
```

```
| examination          |
| information_schema |
| mysql                |
| performance_schema |
| sakila               |
| sys                  |
| world                |
+--------------------+
7 rows in set (0.00 sec)
```

#Command to create a new database in MySQL.

```
mysql> create database streaming_service; Query
OK, 1 row affected (0.01 sec)
```

#Again display the all databases present in the SQL to check if the streaming database is created or not.

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| examination          |
| information_schema |
| mysql                |
| performance_schema |
| sakila               |
| streaming_service  |
| sys                  |
| world                |
+--------------------+
8 rows in set (0.00 sec)
```

#use a streaming dataset.

```
mysql> use streaming_service;
Database changed
```

#Query for creating a table with the col. And their data types.

```
mysql> create table user(user_id int primary key, name varchar(50), email varchar(50), phone
varchar(15), password varchar(50), registration_date date);
Query OK, 0 rows affected (0.04 sec)

mysql> show tables;
+----------------------------+
| Tables_in_streaming_service |
```

```
+----------------------------+
| user                       |
+----------------------------+
1 row in set (0.00 sec)
```

#Query to display the structure of a table, including its column names, data types,and constraints.

```
mysql> desc user;
+------------------+-------------+------+-----+---------+-------+
| Field            | Type        | Null | Key | Default | Extra |
+------------------+-------------+------+-----+---------+-------+
| user_id          | int         | NO   | PRI | NULL    |       |
| name             | varchar(50) | YES  |     | NULL    |       |
| email            | varchar(50) | YES  |     | NULL    |       |
| phone            | varchar(15) | YES  |     | NULL    |       |
| password         | varchar(50) | YES  |     | NULL    |       |
| registration_date | date       | YES  |     | NULL    |       |
+------------------+-------------+------+-----+---------+-------+
6 rows in set (0.01 sec)
```

#Adds a new column address from the user table using alter

```
mysql> alter table user add address varchar(100); Query OK,
0 rows affected (0.16 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Displays the structure of the student table including column names, data types, and keys.

```
mysql> desc user;
+------------------+-------------+------+-----+---------+-------+
| Field            | Type        | Null | Key | Default | Extra |
+------------------+-------------+------+-----+---------+-------+
| user_id          | int         | NO   | PRI | NULL    |       |
| name             | varchar(50) | YES  |     | NULL    |       |
| email            | varchar(50) | YES  |     | NULL    |       |
| phone            | varchar(15) | YES  |     | NULL    |       |
| password         | varchar(50) | YES  |     | NULL    |       |
| registration_date | date       | YES  |     | NULL    |       |
| address          | varchar(100) | YES  |     | NULL    |       |
+------------------+-------------+------+-----+---------+-------+
7 rows in set (0.00 sec)
```

#Drop a column of password from the user table.

```
mysql> alter table user drop column password;
```

Query OK, 0 rows affected (0.04 sec) Records: 0
Duplicates: 0 Warnings: 0

```
mysql> desc user;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| user_id          | int          | NO   | PRI | NULL    |       |
| name             | varchar(50)  | YES  |     | NULL    |       |
| email            | varchar(50)  | YES  |     | NULL    |       |
| phone            | varchar(15)  | YES  |     | NULL    |       |
| registration_date | date        | YES  |     | NULL    |       |
| address          | varchar(100) | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

#Adding a new table for Content.

mysql> create table content(content_id int primary key, title varchar(100), genre varchar(50), language varchar(30), duration int, release_date date, age_rating varchar(10));
Query OK, 0 rows affected (0.04 sec)

#Adding a new table for subscription_plan

mysql> create table subscription_plan(plan_id int primary key, plan_name varchar(30), price int, video_quality varchar(20), max_screens int);
Query OK, 0 rows affected (0.04 sec)

#Adding a new table for subscription.

mysql> create table subscription(subscription_id int primary key, user_id int, plan_id int, start_date date, end_date date, status varchar(20), foreign key(user_id) references user(user_id), foreign key(user_id) references user(user_id), foreign key(plan_id) references subscription_plan(plan_id));
Query OK, 0 rows affected (0.08 sec)

#Adding a new table for payment.

mysql> create table payment(payment_id int primary key, subscription_id int, payment_date date, amount int, payment_method varchar(20), payment_status varchar(20), foreign key(subscription_id) references subscription(subscription_id));
Query OK, 0 rows affected (0.07 sec)

#Adding a new table for watch_history.

mysql> create table watch_history(history_id int primary key, user_id int, content_id int, watch_date date, foreign key(user_id) references user(user_id), foreign key(content_id) references content(content_id));

Query OK, 0 rows affected (0.08 sec)

#Inserting values into user table.

mysql> insert into user
values(1,'Rahul','rahul@gmail.com','9876543210','pass123','2024-01-01'); Query
OK, 1 row affected (0.01 sec)

#Inserting values in the content table.

mysql> insert into content values(101,'Inception','Sci-Fi','English',148,'2010-07-16','PG13'); Query OK, 1 row affected (0.01 sec)

#Inserting values into subscription plan table.

```
mysql> insert into subscription_plan values
    -> (1,'Basic',199,'HD',1),
    -> (2,'Premium',499,'4K',4);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

#Inserting values to subscription table.

```
mysql> insert into subscription values
    -> (1,1,2,'2024-01-01','2024-12-31','Active');
Query OK, 1 row affected (0.01 sec)
```

#Inserting values in payment table.

```
mysql> insert into payment values
    -> (1,1,'2024-01-01',499,'UPI','Success');
Query OK, 1 row affected (0.00 sec)
```

#Inserting values in watch_history table.

```
mysql> insert into watch_history values
    -> (1,1,101,'2024-03-01');
Query OK, 1 row affected (0.01 sec)
```

#Query to display all records from the USER table.

```
mysql> select * from user;
+---------+-------+----------------+------------+----------+------------------+
| user_id | name  | email          | phone      | password | registration_date |
+---------+-------+----------------+------------+----------+------------------+
|       1 | Rahul | rahul@gmail.com | 9876543210 | pass123  | 2024-01-01       |
+---------+-------+----------------+------------+----------+------------------+
1 row in set (0.00 sec)
```

#Query to display all records from the CONTENT table.

```
mysql> select * from content;
+------------+-----------+--------+----------+----------+--------------+------------+
| content_id | title     | genre  | language | duration | release_date | age_rating |
+------------+-----------+--------+----------+----------+--------------+------------+
|        101 | Inception | Sci-Fi | English  |      148 | 2010-07-16   | PG13       |
+------------+-----------+--------+----------+----------+--------------+------------+
1 row in set (0.00 sec)
```

#Query to display all records from the SUBSCRIPTION table.

```
mysql> select * from subscription;
+----------------+---------+---------+------------+------------+--------+
| subscription_id | user_id | plan_id | start_date | end_date   | status |
+----------------+---------+---------+------------+------------+--------+
|              1 |       1 |       2 | 2024-01-01 | 2024-12-31 | Active |
+----------------+---------+---------+------------+------------+--------+
1 row in set (0.00 sec)
```

Theory :

Q1. What is DDL and which DDL commands did you use in your project?

Answer:

DDL (Data Definition Language) is used to define and manage database structure.
In this project, the following DDL commands were used:

- **CREATE** → to create database and tables
- **ALTER** → to modify table structure (add column)
- **DESC** → to display table structure

Q2. What is DML and which DML operations did you perform?

Answer:

DML (Data Manipulation Language) is used to manage data inside tables.
The following DML operations were used:

- **INSERT** → to add data into tables
- **SELECT** → to retrieve data
- **UPDATE** → to modify existing data
- **DELETE** → to remove data

Q3. What is a Primary Key? Give example from your project.

Answer:

A Primary Key is a column that uniquely identifies each record in a table.
It cannot contain NULL or duplicate values.

Example:

In the *students* table, student_id is the primary key.

Q4. Why is ALTER TABLE used?

Answer:

The **ALTER TABLE** command is used to modify the structure of an existing table.
It can be used to add, delete, or modify columns.

Example from project:

ALTER TABLE students ADD phone VARCHAR(15);

This adds a new column phone to the students table.